

Vehicle Claims Management System — Solution Variant B (Event & Evidence Centric)

Premier University — Department of CSE

Semester: Spring 2025 (4th) | Course: CSE 2221 — Database Management Systems | Course Outcome: CO3

Assignment Weight: 10 Marks | Case Study: Red Insurance — Vehicle Claim Form

1. Relational Strategy for the Event/Evidence Model

- **Temporal accuracy:** `PolicySnapshot` and `CaseFile` maintain historical facts as of claim time, something relational databases handle via triggers and transactions.
 - **Complex optional sections:** `VehicleInvolvement`, `WitnessRecord`, and `EvidenceItem` are optional yet linked through foreign keys to keep referential integrity.
 - **Compliance artifacts:** `LawEnforcementAction` records breath/drug tests plus reporting outcomes; relations guarantee they connect to the proper event.
 - **Reporting agility:** SQL views can produce incident timelines or evidence matrices without duplicating information.
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2. Entity Inventory (Conceptual Level)

Entity	Identifier	Core Attributes	Notes
PolicySnapshot	PolicyNum	CustomerRef, ProductLine	Minimal data retained per scope.
CaseFile	CaseID	ClaimFormNumber, ClaimDate, ClaimAmount, ClaimStatus, AtFaultParty	One-to-one with submitted claim; ties to policy and event.

Entity	Identifier	Core Attributes	Notes
AccidentEvent	EventID	EventDate, EventTime, StreetName, CrossStreet, WeatherCondi- tion, RoadSurface, LightingCondi- tion, VehicleParked- Flag	Canonical representation of the accident.
PersonContact	ContactID	FullName, BirthDate, Address, Phone	Shared by drivers, witnesses, and service providers.
DriverProfile	DriverID	LicenceNumber, LicenceStatus, LicenceType, LicenceIssueDate, RelationshipToIn- sured	Reference to PersonContact .
VehicleInvolvement	InvolvementID	RegistrationPlate, Role (In- sured/ThirdParty/Object), MakeModel, InsurerName, PassengerCount	Links a specific vehicle/object to event.
WitnessRecord	WitnessID	StatementSummary, ContactedFlag	Join between event and contact.
EvidenceItem	EvidenceID	EvidenceType (Photo/Quote/Doc), Description, StorageLink	optional attachments per event.
LawEnforcementAction	ActionID	ReportNumber, ReportedToPo- liceFlag, BreathTestRe- sult, DrugTestResult	One row per event capturing policing outcomes.
ServiceProvider	ProviderID	ProviderName, WorkshopAd- dress, ContactID	Distinct table to manage vendor metadata.

Entity	Identifier	Core Attributes	Notes
QuotePackage	QuoteID	QuoteSequence, LaborCost, PartsCost, TotalCost, ApprovedFlag	Each claim requires ≥ 3 quotes; references provider.

3. ERD Narrative (Crow's Foot)

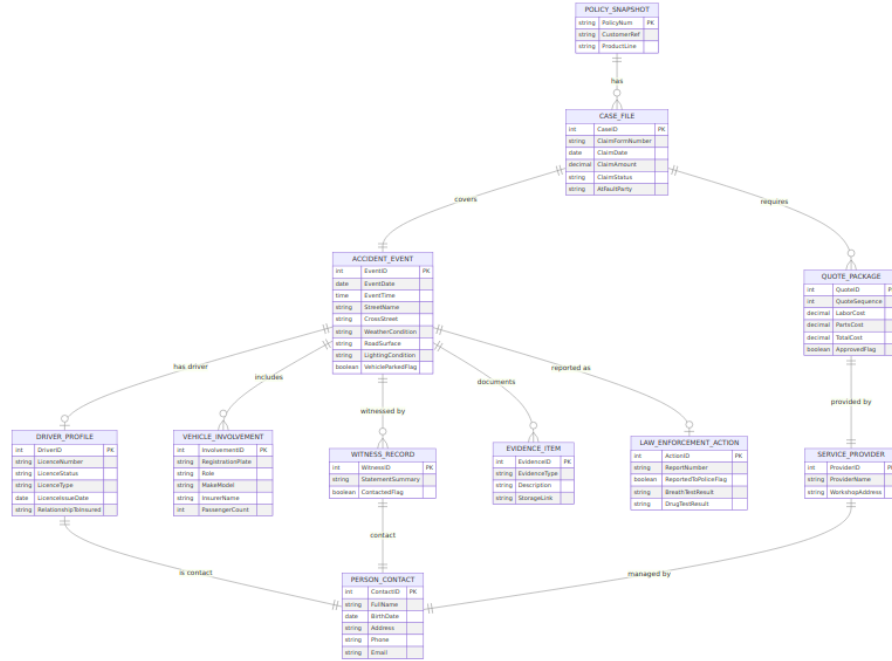


Figure B1: Event & evidence centric ERD.

Relation summary: 1. **PolicySnapshot** (1) $\rightarrow$ (M) **CaseFile**: A policy can have multiple case files over time. 2. **CaseFile** (1) $\rightarrow$ (1) **AccidentEvent**: ensures every claim references exactly one event. 3. **AccidentEvent** (0..1) $\rightarrow$ (1) **DriverProfile** $\rightarrow$ (1) **PersonContact**: driver optional when vehicle parked; driver references a contact record for reuse. 4. **AccidentEvent** (1) $\rightarrow$ (M) **VehicleInvolvement**: includes both insured and other parties, differentiated via Role. 5. **AccidentEvent** (1) $\rightarrow$ (M) **WitnessRecord** $\rightarrow$ (1) **PersonContact**: many witnesses, each tied back to a contact entry. 6. **AccidentEvent** (1) $\rightarrow$ (M) **EvidenceItem**: optional documents, photos, or quotes. 7. **AccidentEvent** (1) $\rightarrow$ (1) **LawEnforcementAction**: each event optionally records police interaction; stored separately for clarity. 8. **CaseFile**

(1) –< (M) **QuotePackage** –(1) **ServiceProvider** –(1) **PersonContact**: ensures vendor reuse and structured quote data. 9. **VehicleInvolvement** rows referencing insured role must map back to the vehicle on the original policy via the **RegistrationPlate**.

4. Normalization Steps

1. **UNF**: Paper form lumps parking info, driver section, police section, quotes, and witnesses with repeating lines (three quotes, unlimited witnesses).
 2. **1NF**: Created atomic tables for quotes, witnesses, evidence; removed repeating groups and structured boolean fields (**VehicleParkedFlag**, **ReportedToPoliceFlag**).
 3. **2NF**: Moved driver licence details to **DriverProfile**, preventing partial dependency on event attributes; **VehicleInvolvement** isolates per-vehicle data rather than repeating within **CaseFile**.
 4. **3NF**: **ServiceProvider** and **PersonContact** remove transitive dependencies (e.g., repairer phone no longer depends on **Quote**). **LawEnforcementAction** holds breath/drug test fields independent of other accident data, ensuring every non-key attribute depends solely on its key.
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5. Relational Model

1. **PolicySnapshot**(<u>PolicyNum</u>, CustomerRef, ProductLine)
2. **CaseFile**(<u>CaseID</u>, ClaimFormNumber, ClaimDate, ClaimAmount, ClaimStatus, AtFaultParty, PolicyNum*, EventID*)
3. **AccidentEvent**(<u>EventID</u>, EventDate, EventTime, StreetName, CrossStreet, WeatherCondition, RoadSurface, LightingCondition, VehicleParkedFlag)
4. **PersonContact**(<u>ContactID</u>, FullName, BirthDate, Address, Phone, Email)
5. **DriverProfile**(<u>DriverID</u>, LicenceNumber, LicenceStatus, LicenceType, LicenceIssueDate, RelationshipToInsured, ContactID*)
6. **VehicleInvolvement**(<u>InvolvementID</u>, RegistrationPlate, Role, MakeModel, InsurerName, PassengerCount, EventID*)

7. `WitnessRecord(<u>WitnessID</u>, StatementSummary, ContactedFlag, EventID*, ContactID*)`
8. `EvidenceItem(<u>EvidenceID</u>, EvidenceType, Description, StorageLink, EventID*)`
9. `LawEnforcementAction(<u>ActionID</u>, ReportNumber, ReportedToPoliceFlag, BreathTestResult, DrugTestResult, EventID*)`
10. `ServiceProvider(<u>ProviderID</u>, ProviderName, WorkshopAddress, ContactID*)`
11. `QuotePackage(<u>QuoteID</u>, QuoteSequence, LaborCost, PartsCost, TotalCost, ApprovedFlag, CaseID*, ProviderID*)`

Integrity notes: `VehicleInvolvement` must contain exactly one row flagged as `Role='Insured'`; `QuotePackage` requires a `CHECK` enforcing positive costs and a trigger to ensure at least three quotes per `CaseID`; `LawEnforcementAction` rows optional but when present must have consistent `ReportNumber`.

6. Design Considerations

- **Event vs Case separation:** Maintains legal audit trails; event data can be shared across reinsurers without exposing financial claim info.
- **Evidence attachments:** `EvidenceItem` future-proofs for photos, police PDFs, or IoT sensor logs.
- **Vendor management:** `ServiceProvider` tied to `PersonContact` so finance can contact the right representative even if workshops rebrand.
- **Vehicle role generalization:** Same table handles insured, other vehicles, and stationary objects, simplifying analytics on collision partners.

7. Responses to Complex Problem Questions

- (a) Requires domain expertise in insurance event modeling and evidence handling.
- (b) Conflicts include balancing compliance requirements (store everything) with privacy (store minimal personal info).
- (c) Converting semi-structured forms into an event/evidence schema demands abstract thinking.

- (d) Rare scenarios (no driver, multiple evidence types) are handled via optional tables.
 - (e) Standards such as ACORD Evidence and ISO police codes inform enumerations.
 - (f) Stakeholders disagree over mandatory evidence uploads vs. speedy claim intake.
 - (g) Evidence, law-enforcement, and quote sub-problems interrelate (e.g., breath test results affecting quote approvals).
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8. Conclusion

- Variant B proves an alternative yet fully compliant relational design emphasising event lineage and evidence governance.
- Future enhancements: add lookup tables for roles/conditions, implement cascading logic for insured vehicle validation, and automate PDF merging of evidence.

Prepared by: [Your Name], Database Management Systems (CSE 2221) — Spring 2025.